Name: _____ Roll No.: _____

EE 2001 Digital Systems Quiz II

Monday, 18/03/2024

Closed book, 50 mins, 20 points

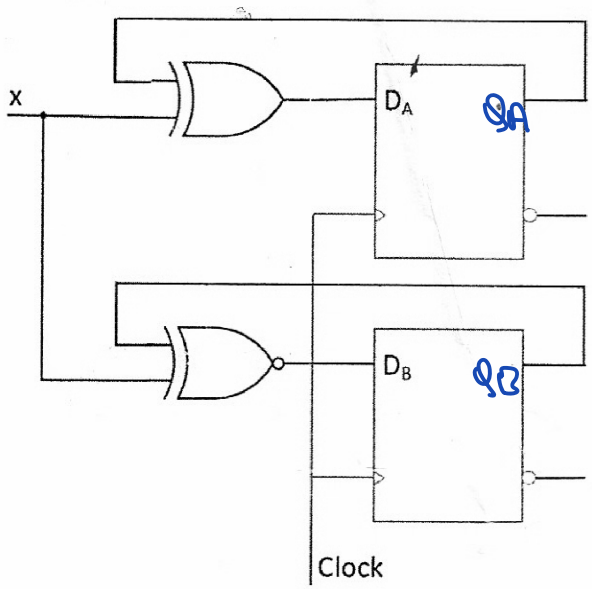
Remember…

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.*

1. Analyze the synchronous sequential circuit provided below and determine the functionality realized by this circuit. (5 pts)

$D_A = Q_A \oplus x$

$D_B = Q_B \odot x$



Q_A	Q_B	x	Q_A^+	Q_B^+
0	0	0	0	1
0	0	1	1	0
0	1	0	0	0
0	1	1	1	1
1	0	0	1	1
1	0	1	0	0
1	1	0	1	0
1	1	1	0	1

2. Design a mod-5 synchronous counter and realize the same using T flip flops. Make sure that any unused state occurrence leads the circuit to return to the initial state i.e., all zeros. (5 pts)

mod 5 : 0 → 1 → 2 → 3 → 4

We need 3 flip flops

Q_C	Q_B	Q_A	Q_C^+	Q_B^+	Q_A^+	T_C	T_B	T_A
0	0	0	0	0	1	0	0	1
0	0	1	0	1	0	0	1	1
0	1	0	0	1	1	0	0	1
0	1	1	1	0	0	1	1	1
1	0	0	0	0	0	1	0	0
1	0	1	0	0	0	1	0	1
1	1	0	0	0	0	1	1	0
1	1	1	0	0	0	1	1	1

Q_C \ Q_BQ_A	00	01	11	10
0			1	
1	1	1	1	1

$T_C = Q_C + Q_B Q_A$

Q_C \ Q_BQ_A	00	01	11	10
0		1	1	
1			1	1

$T_B = Q_A \bar{Q}_C + Q_A Q_B + Q_C Q_B$

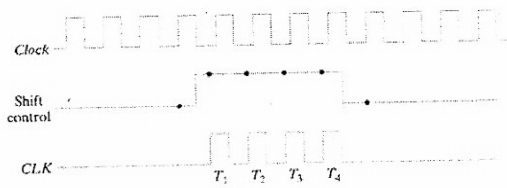
Q_C \ Q_BQ_A	00	01	11	10
0	1	1	1	1
1		1	1	

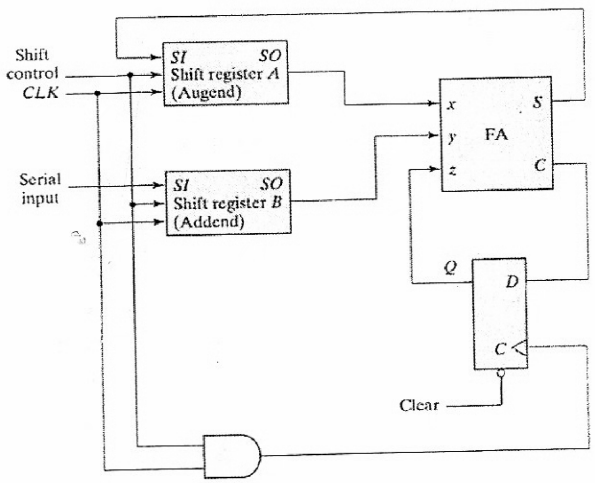
$T_A = \bar{Q}_C + Q_A$

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3. (a) The serial adder of the below figure uses two four-bit registers. Register A holds the binary number 1010 and register B holds 0100. The carry flip-flop is initially reset to 0. List the binary values in register A and the carry flip-flop after every clock cycle.

(2 pts)



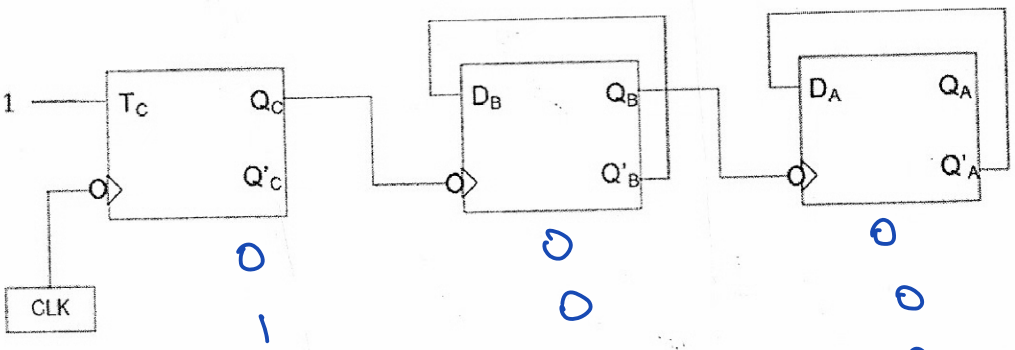


(b) Design a traffic light controller using a 3-bit ring counter such that RED, YELLOW, GREEN lights turn ON sequentially with every clock cycle.

(3 pts)

4. Consider the circuit below. Assume initial states as all 0's. Find the count sequence (Q_A, Q_B, Q_C) and Mod number of the given counter. If we want to reverse the count sequence, what changes are needed in the circuit?

(5 pts)



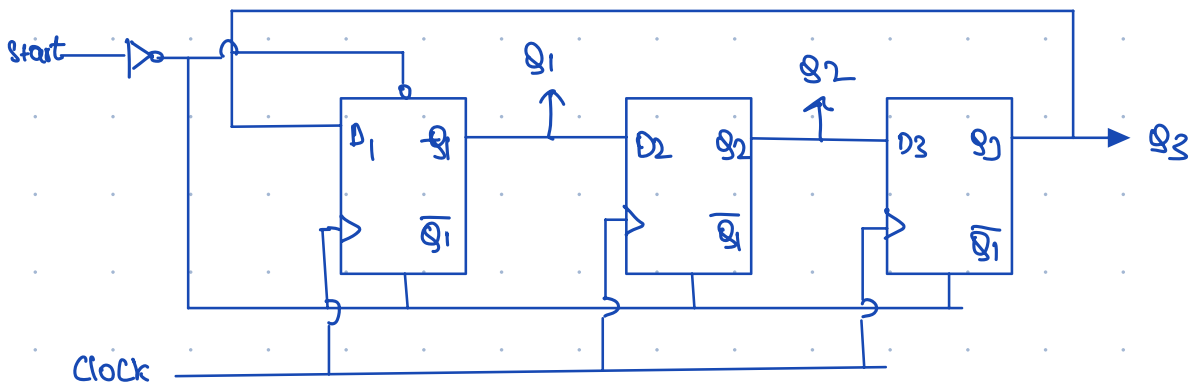
0 0 1 0 1 0

0 0 1 0 1 0

0 0 0 1

mod 8

(b) Design a traffic light controller using a 3-bit ring counter such that RED, YELLOW, GREEN lights turn ON sequentially with every clock cycle.



$Q_1 \rightarrow$ red $Q_2 \rightarrow$ yellow $Q_3 \rightarrow$ green

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EE 2001 Digital Systems Final Exam

Monday, 06/05/2024

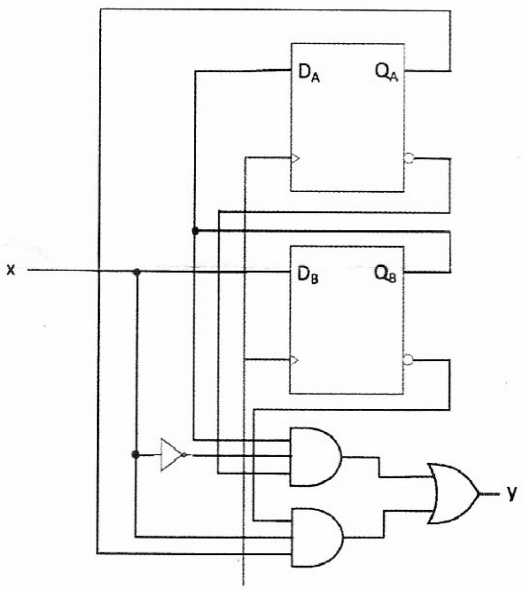
Closed book, 180 mins, 30 points

Remember…

- *Make reasonable assumptions when in doubt. However, you should show all steps and justify assumptions for full credit.*

1. Design and implement a combinational logic circuit that performs a matrix multiplication of a 1x2 matrix consisting of two bit binary numbers and a similar 2x1 matrix. Determine the minimum number of half-adders, full-adders, and other gates required to perform the above task. (5 pts)

2. Analyze the following synchronous sequential circuit with the help of a state table and a corresponding state diagram. Identify whether it is a Mealy or Moore-type circuit and also the operation that it performs. What will be the output for the input sequence 0010101101000 (read from left to right)? (5 pts)



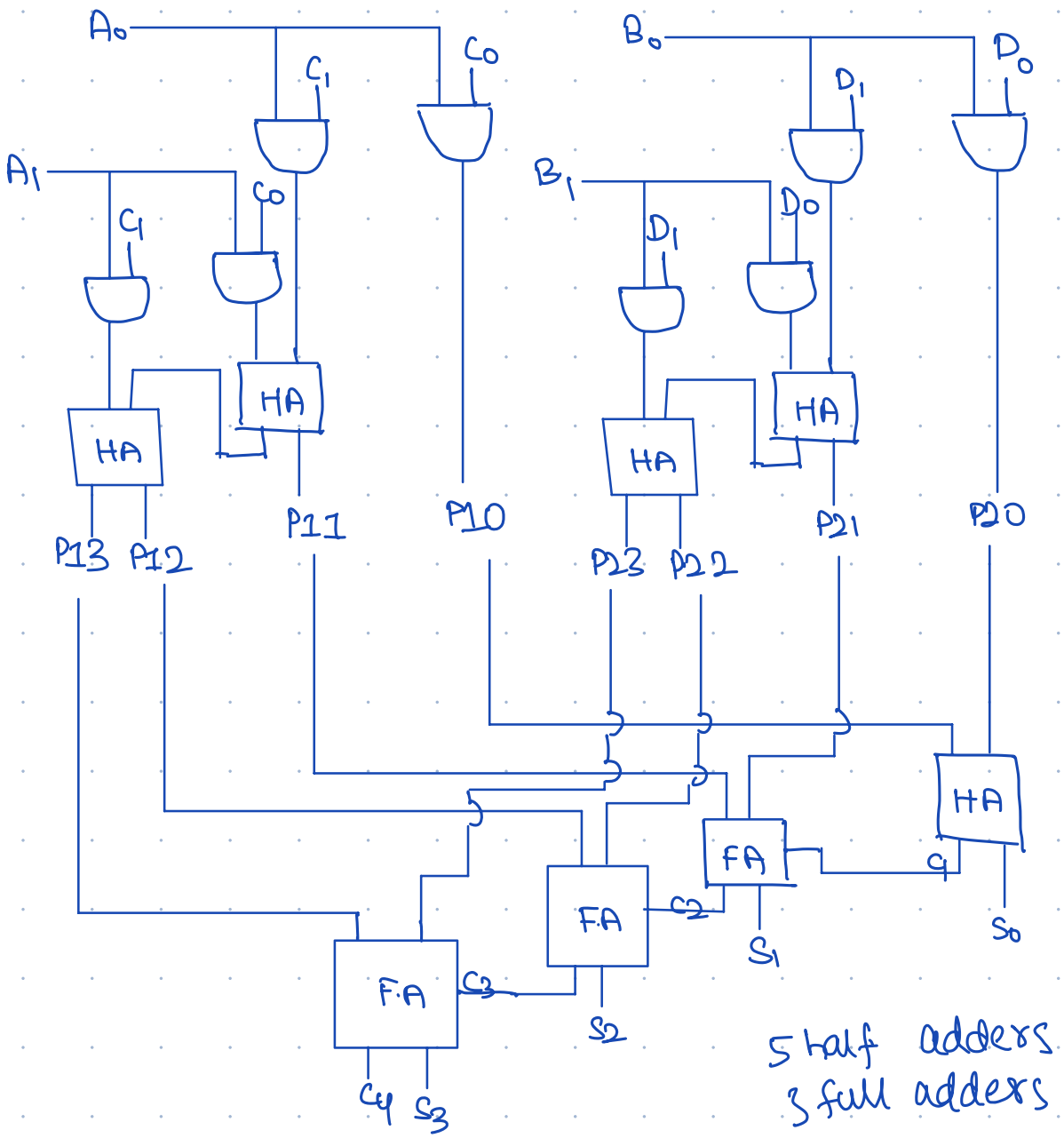
3. Draw the state diagram, state table, and realize a Moore-type error detection circuit for a binary communication link using J-K flip-flops and additional gates, if necessary. When the communication link operates correctly, all blocks of 0's are of odd length and all blocks of 1's are of even length. Whenever an error occurs, i.e. a block of 0's contains an even number of 0's or a block of 1's contain an odd number of 1's, the output Z = 1 immediately with change in input. At all other instances, Z = 0. For example if the input (read from left to right) is 0001100111011110...., the output is 0000000100100000..... (5 pts)

1. Design and implement a combinational logic circuit that performs a matrix multiplication of a 1x2 matrix consisting of two bit binary numbers and a similar 2x1 matrix. Determine the minimum number of half-adders, full-adders, and other gates required to perform the above task. (5 pts)

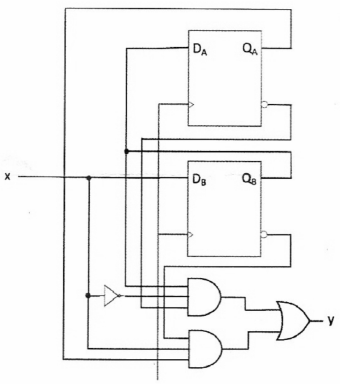
$$\begin{bmatrix} A & B \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} S \end{bmatrix}$$

$A = A_1A_0 \quad C = C_1C_0$

$B = B_1B_0 \quad D = D_1D_0$



2. Analyze the following synchronous sequential circuit with the help of a ~~state table~~ and a corresponding ~~state diagram~~. ~~Identify whether it is a Mealy or Moore-type circuit and~~ also the operation that it performs. What will be the output for the input sequence 0010101101000 (read from left to right)? (5 pts)



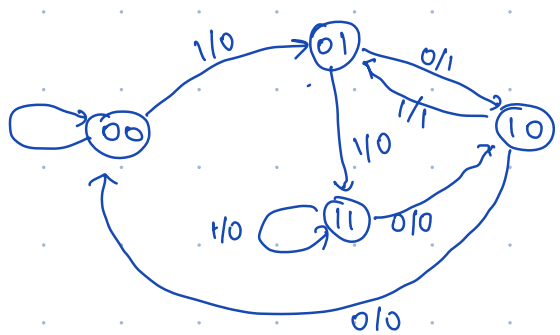
$D_A = Q_B(t) \Rightarrow Q_A(t+1) = Q_B(t)$

$D_B = x \Rightarrow Q_B(t+1) = x$

$Y = Q_B(t)\,\bar{x}\,\overline{Q_A(t)} + \overline{Q_B(t)}\,x\,Q_A(t)$

present states		input	Next States		Output
$Q_A(t)$	$Q_B(t)$	x	$Q_A(t+1)$	$Q_B(t+1)$	Y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
0	1	1	1	1	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	1	0	0
1	1	1	1	1	0

It is a mealy type FSM since the output Y depends on both present state and input 'x'.



Q_A	Q_B	x	Q_A^+	Q_B^+	Y
0	0	0	0	0	0
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	0	1	1
0	1	0	1	0	1
1	0	1	0	1	1
0	1	1	1	1	0
1	1	0	1	0	0
1	0	1	0	1	1
0	1	0	1	0	1
1	0	0	0	0	0
0	0	0	0	0	0

010, 101 Sequence detector

3. Draw the state diagram, state table, and realize a Moore-type error detection circuit for a binary communication link using J-K flip-flops and additional gates, if necessary. When the communication link operates correctly, all blocks of 0's are of odd length and all blocks of 1's are of even length. Whenever an error occurs, i.e. a block of 0's contains an even number of 0's or a block of 1's contain an odd number of 1's, the output Z = 1 immediately with change in input. At all other instances, Z = 0. For example if the input (read from left to right) is 0001100111011110...., the output is 0000000100100000..... (5 pts)

$S_0 \rightarrow$ odd length 0

$S_1 \rightarrow$ even length 0

$S_2 \rightarrow$ Odd length 1

$S_3 \rightarrow$ even length 1

(b) Design a synchronous counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Implement the counter using D flip-flops. (3 pts)

Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	1	0	0
1	0	0	1	1	0
1	1	0	0	0	0

Q_2^+ :

Q_2 \ Q_1Q_0	00	01	11	10
0	0	0	X	1
1	1	X	X	0

$Q_2^+ = \overline{Q_1} Q_2 + \overline{Q_2} Q_1$

Q_1^+ :

Q_2 \ Q_1Q_0	00	01	11	10
0	0	1	X	0
1	1	X	X	0

$Q_1^+ = Q_0 + Q_2 \overline{Q_1}$

Q_0^+ :

Q_2 \ Q_1Q_0	00	01	11	10
0	1	0	X	0
1	0	X	X	0

$Q_0^+ = \overline{Q_2} \overline{Q_1} \overline{Q_0}$

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4. (a) The contents of a four-bit register are initially 1101. The register is shifted six times to the right with the serial input being 1010111. What is the content of the register after each shift? (2 pts)

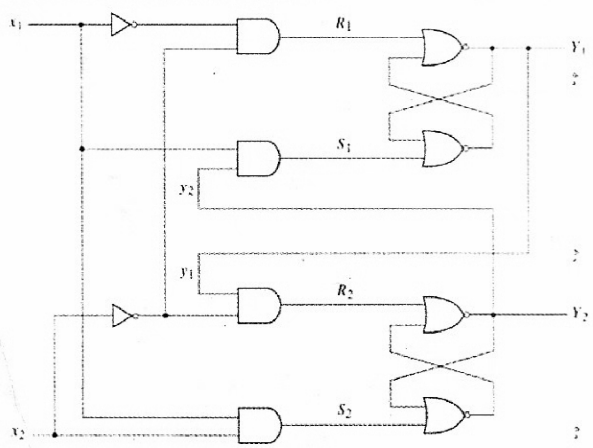
(b) Design a synchronous counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Implement the counter using D flip-flops. (3 pts)

5. (a) Implement the following two Boolean functions using a PLA circuit with minimum number of AND gates and show the corresponding programming table. (2.5 pts)

$$F1(A, B, C) = \sum(0, 1, 2, 4)$$
$$F2(A, B, C) = \sum(0, 5, 6, 7)$$

(b) Assume there are two sensors S1 and S2 in a tank at levels L1 and L2 respectively (L1<L2). There are three lights, Red (R), Orange (O), and Green (G), and a motor control as outputs. The motor should stop after it reaches L2, and the light Red (R) must glow when the water level is below L1, Orange (O) should glow when the water level is between L1 and L2, and Green (G) should glow when the water level is at (or above) L2. Write the truth table for this operation and implement it using PAL. (Note that sensors will be ON when the water level is touching them). (2.5 pts)

6. (a) For the asynchronous sequential circuit shown below, (i) derive the transition table and flow table of the circuit, (ii) Determine if the circuit is stable or not, (iii) Determine all race conditions and check whether they are critical or non-critical. (3 pts)



(b) Draw the logic diagram of the product-of-sums expression

$$Y = (X1 + X2')(X2 + X3)$$

Show that there is a static 0-hazard when X1 and X3 are equal to 0 and X2 goes from 0 to 1. Find a way to remove the hazard. (2 pts)

$Y = (x_1 + \bar{x_2})(x_2 + x_3)$

$x_1 = 0, x_3 = 0, x_2 = 0 \Rightarrow Y = 0$

$x_1 = 0, x_3 = 0, x_2 = 1 \Rightarrow Y = 0$

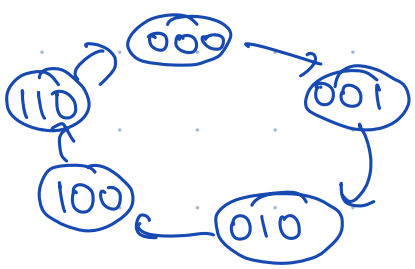
$\bar{x_2}$ doesn't change immediately in reality i.e $x_2 = 1$; $\bar{x_2} = 1$

$\Rightarrow Y = 1$. To remove it, multiply a common term i.e $(x_1 + x_3)$

4. (a) The contents of a four-bit register are initially 1101. The register is shifted six times to the right with the serial input being 1010111. What is the content of the register after each shift?

(b) Design a synchronous counter with the following repeated binary sequence: 0, 1, 2, 4, 6. Implement the counter using D flip-flops. (3 pts)

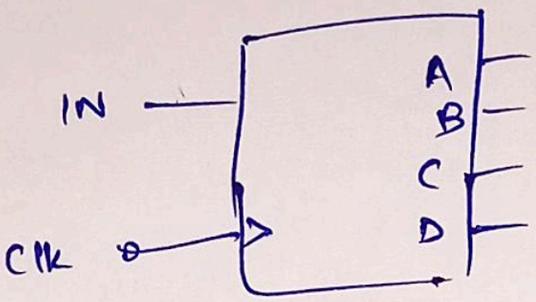
000 → 001 → 010 → 100 → 110



Q_2	Q_1	Q_0	Q_2^+	Q_1^+	Q_0^+
0	0	0	0	0	1
0	0	1	0	1	0
0	1	0	1	0	0
1	0	0	1	1	0
1	1	0	0	0	0

① Implement CARRY and SUM of one-bit full adder using 2×4 DECODERS with active low enable and NAND gates only. Assume that NO complements are available.

② Implement the following chip



such that

A = 1 when even no. of 1, even 0s

B = 1 when odd no. of 1, even 0s

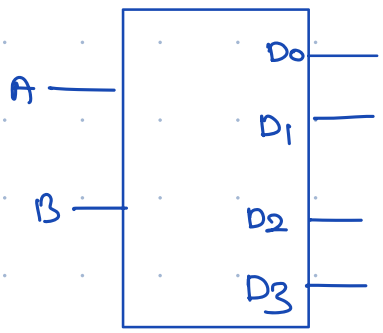
C = 1 when even no. of 1, odd 0s

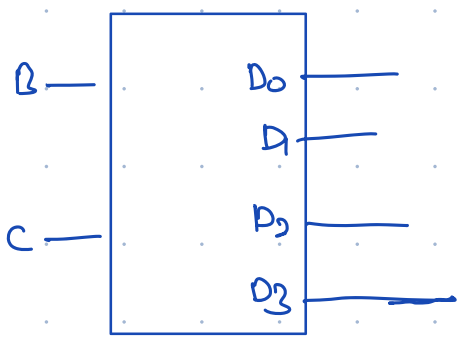
D = 1 when odd no. of 1, odd 0s

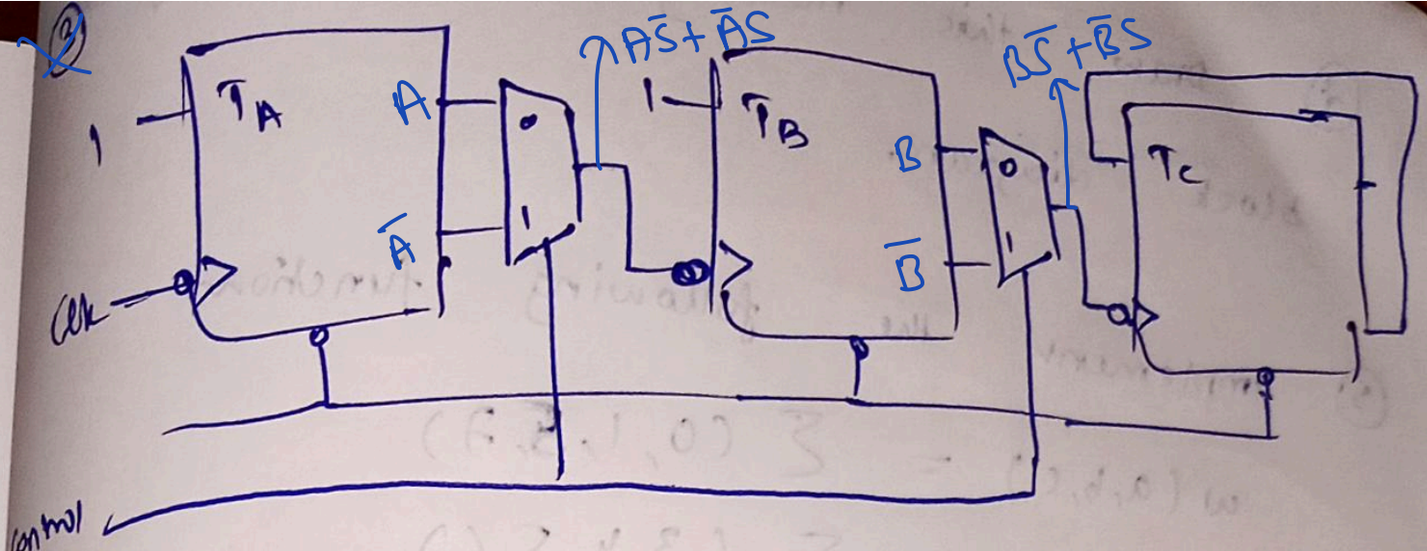
IN is one-bit input stream. Draw the Finite state Machine and realize the same using D-flipflops, state table and an appropriate combinational logic circuit.

① Carry = $AB + BC + AC = \Sigma m(3,5,6,7)$

Sum = $A \oplus B \oplus C = \Sigma m(1,2,4,7)$







Explain the functionality of the circuit in both the cases when control = 0 and 1. Justify the same using a state table for both the cases. Also explain if there are any cycles, ~~states~~ using a state diagram.

(4) (a) How many 32K x 8 RAM chips are needed to create a memory of 512K bytes?

(b) How many address lines are needed to access the 512K bytes? How many of them are connected to all the RAM chips?

(c) What is the size of the decoder needed to access these RAM chips?

(d) Draw this memory circuit as a ~~using blocks~~ block diagram.

(5) Implement the following functions

$$w(a,b,c) = \sum(0,1,5,7)$$

$$x(a,b,c) = \sum(3,4,5,6)$$

$$y(a,b,c) = \sum(1,2,6,7)$$

$$z(a,b,c) = \sum(0,1,2,4)$$

using a PLA. Minimise the expressions using K-Maps.

(6) consider the following asynchronous sequential circuit with internal states y_1, y_2, output z

$$Y_1 = x_1x_2 + x_1'y_2 + x_2'y_1$$

$$Y_2 = x_2 + x_1'y_1 + x_1y_1'y_2$$

$$z = x_2 + y_1$$

(a) Draw the logic diagram.

(b) Derive the transition table and the output maps

(c) Derive the flow table

(d) Comment on stability and race conditions

